Operational Research

5CM522



Module Handbook

Dr. Sam O'Neill

Sensitivity: Internal

Operational Research (5CM522)

Contents

Module Leader 3
Description of the module 3
Relationship to Programme Structure 3
Learning Outcomes 3
Schedule of work 3
Semester 1 4
Semester 2 4
Module Assessment Description 4
Assessment Strategy 4
Component 1 (50%) – Deadline: End of Semester 1 (TBC) 4
Component 2 (50%) - Deadline: End of Semester 2 (TBC) 4

Sensitivity: Internal

Module Leader

- Dr. Sam O'Neill
- s.oneill@derby.ac.uk

Description of the module

The module enables learners to acquire both the ability and confidence to apply Operational Research techniques to a variety of problem areas of importance to business and industry. Emphasis will be placed on applications, for example, forecasting and quality control. The mathematical background will be developed to enhance learners' understanding and enable the development of their modelling abilities.

Commercial software packages will be employed both to enable the solution of realistic problems and to develop learners' experiences of a real life approach to problem solving.

Relationship to Programme Structure

Operational Research (OR) is a fundamental topic in applied mathematics and exposes you to a number of broad areas. A good OR practitioner requires good modelling skills and a good working knowledge of a variety of mathematical and computational tools.

Learning Outcomes

On successful completion of the module, students will be able to:

1. Demonstrate an understanding of Operational Research methods and methodology.
2. Build models to enable the analysis and solution of case study problems.

Schedule of work

This module will be delivered using a blended approach. Each week there will be a 1-hour lecture that you must attend at the time and location indicated on your personalised timetable. You will be allocated to a 1-hour assisted practical session that you are also expected to do tutorial work. This can also be completed outside of the practical/lecture hours.

Below is a breakdown of the module in terms of the hours that are expected.

Activity Type	Hours
Lectures	24
Practicals	24
Guided independent study	152

Scheduled learning and teaching activities	24%
Guided independent study	76%

Sensitivity: Internal

Semester 1

Topic	Week
Introduction to Operations Research	1
Linear Programming 1	2
Linear Programming 2	3
Mathematical Programming – Modelling and Some Issues	4
Mixed Integer Programming	5
Integer Linear Programming – Further Problems	6
Scheduling	7
Network Models: Transportation and Transhipment Problem	8
Travelling Salesman Problem	9
Vehicle Routing Problem	10
Inventory Modelling	11
Review	12

Semester 2

Topic	Week
Introduction to Simulation	1
Queuing Theory	2
Discrete-Event Simulation	3
Agent Based Modelling	4
Coursework 2 Release	5
The Beer Game	6
From Beer Game to Systems Dynamic Modelling	7
Forecasting	8
Times Series	9
Decisions under Uncertainty: Robust and Stochastic Linear Programming	10
Predict Then Optimise: ML + Optimisation	11
In-class Test Revision	12

Module Assessment Description

The assessment is made up of two components each worth 50% of the module.

- Coursework (50%).
- Timed-based In-class Test (50%).

Full details of these assessments are available by the assignment brief documents. There will also be additional information given to you in the lectures and practicals.

Assessment Strategy

Component 1 (50%)

Students are required to model and answer a series of questions related to Operational Research. See individual assignment briefs for details.

Component 2 (50%)

Students are required to sit a timed-based in-class test in which they will be presented with a series of problems related to Operational Research.

Sensitivity: Internal